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Domain :- AI - ML

Branch Name :- Computer Engineering

<https://github.com/pratikshatonde12-dotcom/Python-Basic-Projects..git>

- **Countries Dataset – Data Reading, Loading, Exploration, and Understanding.**

➤ **Aim**

To read, load, explore, and understand the Countries dataset using the Pandas library in Python.

➤ **Objective**

To explain the basic steps of loading a dataset and performing exploratory data analysis (EDA) using commonly used Pandas functions.

1) Import the Pandas library.

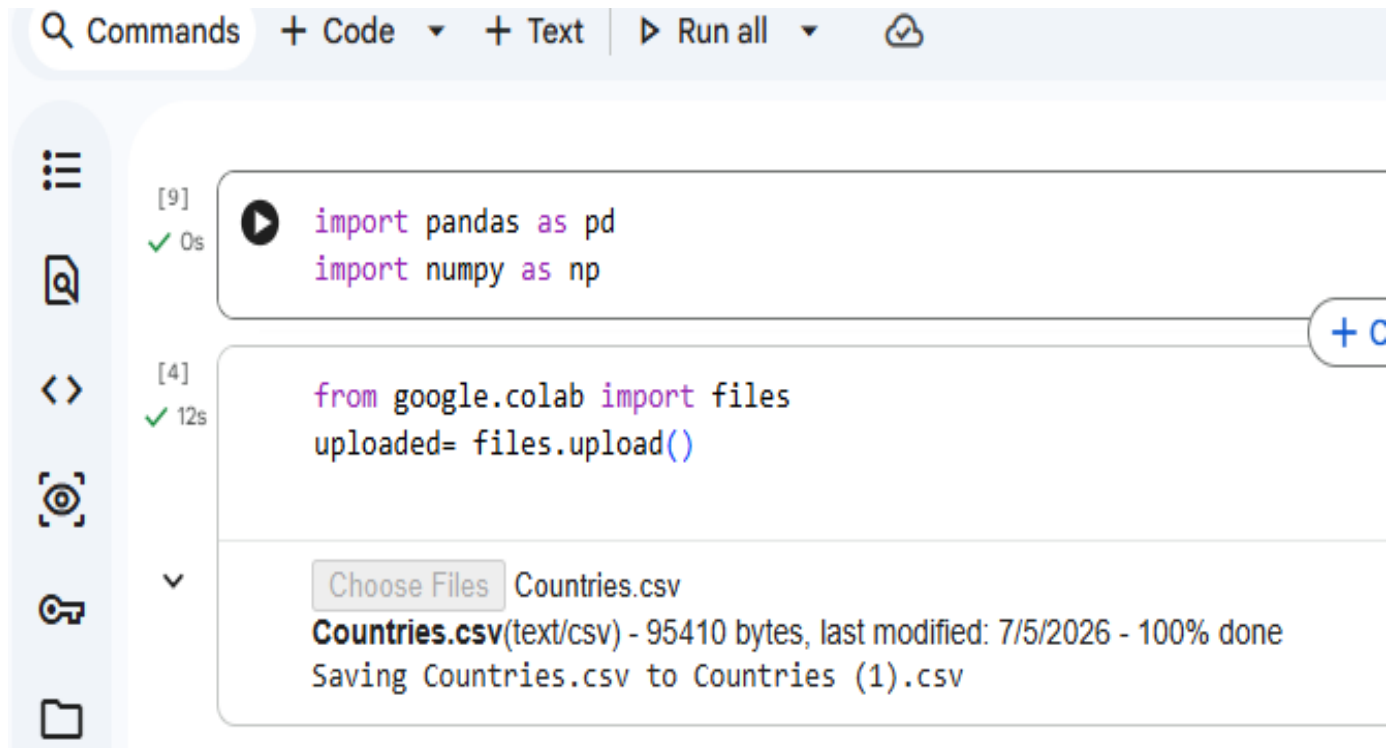
The screenshot shows a Google Colab notebook interface. The browser tabs at the top include 'Inbox (94) - pratikshatonde12@', 'Untitled9.ipynb - Colab', and 'Classwork for AIML Development'. The address bar shows the URL 'colab.research.google.com/drive/1Uau5MWzKCKToKCfYu0GwEL5labLk2nh#scrollTo=VhT5xaMBMKjP'. The notebook toolbar shows 'Commands', '+ Code', '+ Text', and 'Run all'. The left sidebar contains icons for file management, search, and other tools. The main code area displays two code cells. The first cell, labeled '[9]', contains the code 'import pandas as pd' and 'import numpy as np', with a green checkmark and '0s' indicating successful execution. The second cell, labeled '[4]', contains the code 'from google.colab import files' and 'uploaded= files.upload()', with a green checkmark and '12s' indicating successful execution. Below the second cell, a file upload dialog is shown, displaying 'Choose Files' and 'Countries.csv'. The file details are: 'Countries.csv(text/csv) - 95410 bytes, last modified: 7/5/2026 - 100% done'. The status 'Saving Countries.csv to Countries (1).csv' is also visible.

```
[9] import pandas as pd
import numpy as np
```

```
[4] from google.colab import files
uploaded= files.upload()
```

Choose Files Countries.csv
Countries.csv(text/csv) - 95410 bytes, last modified: 7/5/2026 - 100% done
Saving Countries.csv to Countries (1).csv

2) Load the CSV dataset into a DataFrame.



The image shows a Google Colab interface. At the top, there is a toolbar with a search icon, the text 'Commands', and buttons for '+ Code', '+ Text', 'Run all', and a cloud icon. On the left side, there is a vertical sidebar with icons for a menu, search, code editor, output, file explorer, and a folder icon. The main area displays two code cells. The first cell, labeled '[9]' with a green checkmark and '0s', contains the code: `import pandas as pd` and `import numpy as np`. The second cell, labeled '[4]' with a green checkmark and '12s', contains the code: `from google.colab import files` and `uploaded= files.upload()`. Below the second cell, there is a file upload section showing a 'Choose Files' button, the filename 'Countries.csv', and the status: 'Countries.csv(text/csv) - 95410 bytes, last modified: 7/5/2026 - 100% done' and 'Saving Countries.csv to Countries (1).csv'. A '+ C' button is visible on the right side of the first code cell.

Commands + Code + Text ▶ Run all

[9] ✓ 0s

```
import pandas as pd
import numpy as np
```

[4] ✓ 12s

```
from google.colab import files
uploaded= files.upload()
```

Choose Files Countries.csv

Countries.csv(text/csv) - 95410 bytes, last modified: 7/5/2026 - 100% done
Saving Countries.csv to Countries (1).csv

3) Display the all dataset.

df

	country	country_long	currency	capital_city	region	continent	demonym	latitude	longitude	agricultural_land	...	population	women
0	Afghanistan	Islamic State of Afghanistan	Afghan afghani	Kabul	Southern Asia	Asia	Afghan	33.000000	65.000000	383560.0	...	41128771	
1	Albania	Republic of Albania	Albanian lek	Tirana	Southern Europe	Europe	Albanian	41.000000	20.000000	11655.5	...	2775634	
2	Algeria	People's Democratic Republic of Algeria	Algerian dinar	Algiers	Northern Africa	Africa	Algerian	28.000000	3.000000	413588.0	...	44903225	
3	Andorra	Principality of Andorra	Euro	Andorra la Vella	Southern Europe	Europe	Andorran	42.500000	1.500000	187.2	...	79824	
4	Angola	People's Republic of Angola	Angolan kwanza	Luanda	Middle Africa	Africa	Angolan	-12.500000	18.500000	569525.0	...	35588987	
...	...	...	...	...	...	...	...	...	...	...	...	...	...
189	Vietnam	Socialist Republic of Vietnam	Vietnamese dong	Hanoi	South-Eastern Asia	Asia	Vietnamese	16.166667	107.833333	123600.0	...	98186856	
190	West Bank	West Bank	Israeli new sheqel	Ramallah	Western Asia	Asia	Israeli	31.000000	35.000000	4440.0	...	5042640	

Terminal 3:50 PM Python 3

4) Display the all columns in dataset.

13]
0s



df.columns

✓

```
*** Index(['country', 'country_long', 'currency', 'capital_city', 'region',
          'continent', 'demonym', 'latitude', 'longitude', 'agricultural_land',
          'forest_area', 'land_area', 'rural_land', 'urban_land',
          'central_government_debt_pct_gdp', 'expense_pct_gdp', 'gdp',
          'inflation', 'self_employed_pct', 'tax_revenue_pct_gdp',
          'unemployment_pct', 'vulnerable_employment_pct',
          'electricity_access_pct', 'alternative_nuclear_energy_pct',
          'electricity_production_coal_pct',
          'electricity_production_hydroelectric_pct',
          'electricity_production_gas_pct', 'electricity_production_nuclear_pct',
          'electricity_production_oil_pct', 'electricity_production_renewable_pct',
          'energy_imports_pct', 'fossil_energy_consumption_pct',
          'renewable_energy_consumption_pct', 'co2_emissions',
          'methane_emissions', 'nitrous_oxide_emissions',
          'greenhouse_other_emissions', 'urban_population_under_5m',
          'health_expenditure_pct_gdp', 'health_expenditure_capita',
          'hospital_beds', 'hiv_incidence', 'suicide_rate', 'armed_forces',
          'internally_displaced_persons', 'military_expenditure_pct_gdp',
          'birth_rate', 'death_rate', 'fertility_rate', 'internet_pct',
          'life_expectancy', 'net_migration', 'population_female',
          'population_male', 'population', 'women_parliament_seats_pct',
          'rural_population', 'urban_population', 'press', 'democracy_score',
          'democracy_type', 'median_age', 'political_leader', 'title'],
          dtype='object')
```

5) Display the first 5 rows of the dataset.

df.head()

	country	country_long	currency	capital_city	region	continent	demonym	latitude	longitude	agricultural_land	...	population	women_parliame
0	Afghanistan	Islamic State of Afghanistan	Afghan afghani	Kabul	Southern Asia	Asia	Afghan	33.0	65.0	383560.0	...	41128771	
1	Albania	Republic of Albania	Albanian lek	Tirana	Southern Europe	Europe	Albanian	41.0	20.0	11655.5	...	2775634	
2	Algeria	People's Democratic Republic of Algeria	Algerian dinar	Algiers	Northern Africa	Africa	Algerian	28.0	3.0	413588.0	...	44903225	
3	Andorra	Principality of Andorra	Euro	Andorra la Vella	Southern Europe	Europe	Andorran	42.5	1.5	187.2	...	79824	
4	Angola	People's Republic of Angola	Angolan kwanza	Luanda	Middle Africa	Africa	Angolan	-12.5	18.5	569525.0	...	35588987	

5 rows x 64 columns

6) Display the last 5 rows of the dataset.

df.tail()

	country	country_long	currency	capital_city	region	continent	demonym	latitude	longitude	agricultural_land	...	population	women_p
189	Vietnam	Socialist Republic of Vietnam	Vietnamese dong	Hanoi	South-Eastern Asia	Asia	Vietnamese	16.166667	107.833333	123600.0	...	98186856	
190	West Bank and Gaza	West Bank and Gaza	Israeli new shekel	Ramallah	Western Asia	Asia	Palestinian	31.900000	35.200000	4449.0	...	5043612	
191	Yemen	Republic of Yemen	Yemeni rial	Sana'a	Western Asia	Asia	Yemeni	15.000000	48.000000	234520.0	...	33696614	
192	Zambia	Republic of Zambia	New Zambian kwacha	Lusaka	Eastern Africa	Africa	Zambian	-15.000000	30.000000	238360.0	...	20017675	
193	Zimbabwe	Republic of Zimbabwe	U.S. dollar	Harare	Eastern Africa	Africa	Zimbabwean	-20.000000	30.000000	162000.0	...	16320537	

5 rows x 64 columns

7) Show the number of rows and columns.

[17]
✓ Os



`df.shape`



`(194, 64)`

8) Show dataset information, data types, and non-null values.

```
[18] df.info()
✓ Os
```

8	longitude	194	non-null	float64
9	agricultural_land	193	non-null	float64
10	forest_area	194	non-null	float64
11	land_area	194	non-null	float64
12	rural_land	194	non-null	float64
13	urban_land	194	non-null	float64
14	central_government_debt_pct_gdp	120	non-null	float64
15	expense_pct_gdp	156	non-null	float64
16	gdp	193	non-null	float64
17	inflation	184	non-null	float64
18	self_employed_pct	179	non-null	float64
19	tax_revenue_pct_gdp	159	non-null	float64
20	unemployment_pct	179	non-null	float64
21	vulnerable_employment_pct	179	non-null	float64
22	electricity_access_pct	194	non-null	float64
23	alternative_nuclear_energy_pct	138	non-null	float64
24	electricity_production_coal_pct	138	non-null	float64
25	electricity_production_hydroelectric_pct	138	non-null	float64
26	electricity_production_gas_pct	138	non-null	float64
27	electricity_production_nuclear_pct	138	non-null	float64
28	electricity_production_oil_pct	138	non-null	float64
29	electricity_production_renewable_pct	138	non-null	float64
30	energy_imports_pct	138	non-null	float64
31	fossil_energy_consumption_pct	168	non-null	float64
32	renewable_energy_consumption_pct	192	non-null	float64
33	co2_emissions	191	non-null	float64
34	methane_emissions	191	non-null	float64
35	nitrous_oxide_emissions	191	non-null	float64

```

... 36 greenhouse_other_emissions      184 non-null float64
    37 urban_population_under_5m      194 non-null float64
    38 health_expenditure_pct_gdp      190 non-null float64
    39 health_expenditure_capita        190 non-null float64
    40 hospital_beds                    192 non-null float64
    41 hiv_incidence                    133 non-null float64
    42 suicide_rate                     183 non-null float64
    43 armed_forces                     174 non-null float64
    44 internally_displaced_persons      73 non-null float64
    45 military_expenditure_pct_gdp      163 non-null float64
    46 birth_rate                       194 non-null float64
    47 death_rate                       194 non-null float64
    48 fertility_rate                   193 non-null float64
    49 internet_pct                     194 non-null float64
    50 life_expectancy                  191 non-null float64
    51 net_migration                     194 non-null int64
    52 population_female                 194 non-null int64
    53 population_male                   194 non-null int64
    54 population                         194 non-null int64
    55 women_parliament_seats_pct        193 non-null float64
    56 rural_population                  194 non-null int64
    57 urban_population                  194 non-null int64
    58 press                             194 non-null float64
    59 democracy_score                   194 non-null float64
    60 democracy_type                     194 non-null object
    61 median_age                         194 non-null float64
    62 political_leader                  187 non-null object
    63 title                             187 non-null object
dtypes: float64(48), int64(6), object(10)
memory usage: 97.1+ KB

```

9) Generate summary statistics for numerical columns.

9] 0s df.describe()

***	latitude	longitude	agricultural_land	forest_area	land_area	rural_land	urban_land	central_government_debt_pct_gdp	expense_pct_gdp
count	194.000000	194.000000	1.930000e+02	1.940000e+02	1.940000e+02	1.940000e+02	194.000000	120.000000	156.000000
mean	18.975601	22.027491	2.454551e+05	2.086784e+05	6.675087e+05	6.563711e+05	9777.116531	66.759366	30.051403
std	23.876225	66.396389	6.356268e+05	7.824926e+05	1.837107e+06	1.811169e+06	42301.458421	71.806247	26.740880
min	-41.000000	-175.000000	4.000000e+00	0.000000e+00	2.027000e+00	3.495450e-02	0.000000	0.000000	0.000267
25%	4.000000	-5.000000	6.464000e+03	3.331775e+03	2.355250e+04	2.186562e+04	359.618250	31.951300	18.371875
50%	16.583333	21.500000	3.872780e+04	2.528925e+04	1.203750e+05	1.159945e+05	1645.170000	55.426850	27.337750
75%	40.000000	50.162500	2.150000e+05	1.236735e+05	5.237000e+05	4.911505e+05	4054.022500	79.539300	35.083500
max	65.000000	178.000000	5.285080e+06	8.153120e+06	1.637690e+07	1.622420e+07	522345.000000	687.994000	310.443000

8 rows x 54 columns

10) Display the data type of each column.

[20]
✓ 0s



`df.dtypes`

...

0

country

object

country_long

object

currency

object

capital_city

object

region

object

...

...

democracy_score

float64

democracy_type

object

median_age

float64

political_leader

object

title

object

64 rows × 1 columns

dtype: object

11) Count missing values in each column.

```
[21]  
✓ Os  
df.isnull().sum()  
country      0  
country_long  0  
currency      0  
capital_city  0  
region        0  
...          ...  
democracy_score  0  
democracy_type  0  
median_age      0  
political_leader  7  
title           7  
64 rows × 1 columns  
dtype: int64
```

12) Count unique values in each column.

[22]
✓ 0s



```
df.nunique()
```



...

0

country

194

country_long

194

currency

142

capital_city

194

region

22

...

...

democracy_score

146

democracy_type

5

median_age

143

political_leader

187

title

17

64 rows × 1 columns

dtype: int64

13) Display 5 random rows from the dataset.

[23] ✓ 0s df.sample(5)

	country	country_long	currency	capital_city	region	continent	demonym	latitude	longitude	agricultural_land	...	population	women_pa
38	Costa Rica	Republic of Costa Rica	Costa Rican colon	San José	Central America	Americas	Costa Rican	10.000000	-84.000000	17615.0	...	5180829	
178	Turkmenistan	Turkmenistan	New Turkmen manat	Ashgabat	Central Asia	Asia	Turkmen	40.000000	60.000000	338380.0	...	6430770	
52	El Salvador	Republic of El Salvador	U.S. dollar	San Salvador	Central America	Americas	Salvadoran	13.833333	-88.916667	11957.0	...	6336392	
61	Gabon	Gabonese Republic	Central African CFA franc	Libreville	Middle Africa	Africa	Gabonese	-1.000000	11.750000	22126.4	...	2388992	
1	Albania	Republic of Albania	Albanian lek	Tirana	Southern Europe	Europe	Albanian	41.000000	20.000000	11655.5	...	2775634	

5 rows x 64 columns

14) Count occurrences of unique values (for a selected column).

df.value_counts()

country	country_long	currency	capital_city	region	continent	demonym	latitude	longitude	agricultural_land	forest_area	land_area
Azerbaijan	Republic of Azerbaijan	New Azeri manat	Baku	Western Asia	Asia	Azerbaijani	40.500000	47.500000	47801.00	11317.7	82646.0
Bangladesh	People's Republic of Bangladesh	Bangladeshi taka	Dhaka	Southern Asia	Asia	Bangladeshi	24.000000	90.000000	99010.00	18834.0	130170.0
Bolivia	Plurinational State of Bolivia	Bolivian Boliviano	Sucre	South America	Americas	Bolivian	-17.000000	-65.000000	377870.00	508338.0	1083300
Brazil	Federative Republic of Brazil	Brazilian real	Brasília	South America	Americas	Brazilian	-10.000000	-55.000000	2368790.00	4966200.0	8358140
Cameroon	Republic of Cameroon	Central African CFA franc	Yaoundé	Middle Africa	Africa	Cameroonian	6.000000	12.000000	97500.00	203405.0	472710.0
Colombia	Republic of Colombia	Colombian peso	Bogotá	South America	Americas	Colombian	4.000000	-72.000000	482428.00	591419.0	1109500
Congo	Republic of Congo	Central African CFA franc	Brazzaville	Middle Africa	Africa	Congolese	-1.000000	15.000000	106280.00	219460.0	341500.0

variables Terminal 4:19 PM Python 3

...	Croatia	Republic of Croatia	Croatian kuna	Zagreb	Southern Europe	Europe	Croatian	45.166667	15.500000	15050.00		
	Cyprus	Republic of Cyprus	Euro	Nicosia	Southern Europe	Europe	Cypriot	35.000000	33.000000	1341.29	1725.3	9240.0
	Côte d'Ivoire	Republic of Côte d'Ivoire	West African CFA franc	Yamoussoukro	Western Africa	Africa	Ivorian	8.000000	-5.000000	212000.00	28367.1	318000.
	Dem. Rep. Congo	Democratic Republic of the Congo	Congolese franc	Kinshasa	Middle Africa	Africa	Congolese	0.000000	25.000000	335720.00	1261550.0	2267050
	El Salvador	Republic of El Salvador	U.S. dollar	San Salvador	Central America	Americas	Salvadoran	13.833333	-88.916667	11957.00	5838.8	20720.0
	Ethiopia	Federal Democratic Republic of Ethiopia	Ethiopian birr	Addis Ababa	Eastern Africa	Africa	Ethiopian	8.000000	38.000000	384761.00	170685.0	1128570
	Georgia	Georgia	Georgian lari	Tbilisi	Western Asia	Asia	Georgian	42.000000	43.500000	23779.00	28224.0	69490.0
	Guatemala	Republic of Guatemala	Guatemalan quetzal	Guatemala City	Central America	Americas	Guatemalan	15.500000	-90.250000	38560.00	35278.0	107160.
	India	Republic of India	Indian rupee	New Delhi	Southern Asia	Asia	Indian	20.000000	77.000000	1790450.00	721600.0	2973190

...	Indonesia	Republic of Indonesia	Indonesian rupiah	Jakarta	South-Eastern Asia	Asia	Indonesian	-5.000000	120.000000	623000.00		
	Kazakhstan	Republic of Kazakhstan	Kazakh tenge	Astana	Central Asia	Asia	Kazakhstani	48.000000	68.000000	2140030.00	34546.8	2699700
	Kyrgyz Republic	Kyrgyz Republic	Kyrgyz som	Bishkek	Central Asia	Asia	Kirghiz	41.000000	75.000000	103678.00	13153.8	191800.
	Lebanon	Lebanese Republic	Lebanese pound	Beirut	Western Asia	Asia	Lebanese	33.833333	35.833333	6696.70	1433.3	10230.0
	Mexico	United Mexican States	Mexican peso	Mexico City	Central America	Americas	Mexican	23.000000	-102.000000	971380.00	656921.0	1943950
	Myanmar	Republic of the Union of Myanmar	Myanmar kyat	Naypyidaw	South-Eastern Asia	Asia	Burmese	22.000000	98.000000	130080.00	285439.0	652670.
	Peru	Republic of Peru	Peruvian new sol	Lima	South America	Americas	Peruvian	-10.000000	-76.000000	244783.00	723304.0	1280000

...

Philippines	Republic of the Philippines	Philippine peso	Manila	South-Eastern Asia	Asia	Filipino	13.000000	122.000000	126750.00		
Sri Lanka	Democratic Socialist Republic of Sri Lanka	Sri Lankan rupee	Colombo	Southern Asia	Asia	Sri Lankan	7.000000	81.000000	28120.00	21130.2	61860.0
Sudan	Republic of the Sudan	Sudanese pound	Khartoum	Northern Africa	Africa	Sudanese	15.000000	30.000000	694058.00	183596.0	1868000
Thailand	Kingdom of Thailand	Thai baht	Bangkok	South-Eastern Asia	Asia	Thai	15.000000	100.000000	230100.00	198730.0	510890.0
Ukraine	Ukraine	Ukrainian hryvnia	Kiev	Eastern Europe	Europe	Ukrainian	49.000000	32.000000	413110.00	96900.0	579400.0

dtype: int64